==========

APEX RETAIL GROUP | ANALYTICS TRAINING PROGRAM
DOCUMENT: CLASS NOTES (SESSION 6)
TOPIC: SLICING & GROUPING DATA — UNCOVERING PERFORMANCE DRIVERS
LEVEL: INTERMEDIATE (DAY 6)

==========

1. CONTINUITY & BUSINESS SCENARIO RECAP

In Session 5, you learned how to clean messy datasets by handling missing values, removing duplicates, standardizing text strings, and parsing dates. You left the customer returns data clean, structured, and audit-ready.

Now, in your sixth week at Apex Retail Group:

Marcus Thorne, Product Category Manager, reaches out with a puzzling business problem:

> **"Our overall store revenue figures look stable on paper, but executive leadership suspects certain product lines are silently bleeding profits while others carry the entire catalog.**
>
> **I have our clean quarterly transaction dataset. I need you to break down performance by product category, identify our top-selling lines versus profit detractors, evaluate purchase frequency across customer tiers, and segment our product portfolio.**
>
> **Don't just give me summary totals—show me where the revenue and profit drivers are hiding."**

In this session, you step beyond basic DataFrame inspection and cleaning. You will learn how to group, aggregate, sort, and segment tabular data using Pandas `` `.groupby()` `` and multi-metric aggregation methods.

2. FROM CLEAN DATA TO BUSINESS SEGMENTATION

Cleaning data (Session 5) ensures our records are accurate. But a clean dataset alone does not answer stakeholder questions.

To uncover performance drivers, analysts use **Aggregation and Segmentation**:
- **Aggregation**: Condensing thousands of individual transactions into category-level summaries (e.g., total sales, average margin).
- **Segmentation**: Splitting data into meaningful business groups (e.g., product categories, customer tiers, geographical regions) to compare performance patterns.

3. PANDAS GROUPING & AGGREGATION TOOLKIT

A. The Split-Apply-Combine Pattern

Grouping in Pandas follows three logical steps:
1. **Split**: Divide the dataset into groups based on a categorical column (e.g., `product_category`).
2. **Apply**: Compute a mathematical function on each group (e.g., sum revenue, mean price).
3. **Combine**: Merge the results into a clean summary DataFrame.

B. Basic Grouping & Single Aggregation (`.groupby()`)

To calculate total sales by product category:

```
category_sales = df.groupby("product_category")["revenue"].sum()
```

C. Multi-Metric Aggregation (`.agg()`)
To calculate multiple statistics simultaneously (e.g., total revenue, average order value, and total quantity sold) for each category:

```
category_summary = df.groupby("product_category").agg({
    "revenue": ["sum", "mean"],
    "quantity": "sum"
})
```

D. Sorting & Ranking (`sort_values()`)
To isolate top or bottom performers, analysts sort aggregated outputs:

```
top_categories = category_summary.sort_values(by=("revenue", "sum"), ascending=False)
```

E. Frequency Distribution (`value_counts()`)
To count occurrences of categorical items (e.g., number of orders per customer tier):

```
tier_counts = df["customer_tier"].value_counts()
```

4. EXECUTING THE CATEGORY PROFITABILITY ANALYSIS IN PANDAS

Below is the clean, annotated Python code required for Marcus's product category audit:

```python

import pandas as pd

# Step 1: Load clean quarterly transaction dataset
data = {
    "order_id": ["ORD-601", "ORD-602", "ORD-603", "ORD-604", "ORD-605", "ORD-606", "ORD-607", "ORD-608", "ORD-609", "ORD-610"],
    "product_category": ["Electronics", "Apparel", "Home", "Electronics", "Apparel", "Home", "Electronics", "Home", "Apparel", "Electronics"],
    "customer_tier": ["Gold", "Silver", "Bronze", "Gold", "Silver", "Gold", "Bronze", "Silver", "Bronze", "Gold"],
    "units_sold": [2, 5, 1, 3, 4, 2, 1, 6, 3, 2],
    "revenue": [1200.00, 250.00, 80.00, 1800.00, 200.00, 160.00, 600.00, 480.00, 150.00, 1200.00],
    "profit_margin": [0.25, 0.45, 0.10, 0.30, 0.40, 0.15, 0.28, 0.12, 0.50, 0.25]
}

df = pd.DataFrame(data)

# Step 2: Group by product category and calculate total revenue and total units sold
category_volume = df.groupby("product_category")[["revenue", "units_sold"]].sum()

print("--- TOTAL REVENUE & UNITS BY CATEGORY ---")
print(category_volume)

# Step 3: Multi-metric aggregation (Sum of revenue, Mean of profit margin)
category_performance = df.groupby("product_category").agg({
    "revenue": "sum",
    "profit_margin": "mean",
    "order_id": "count"
}).rename(columns={"order_id": "order_count"})
```

```
print("\n--- CATEGORY PERFORMANCE SUMMARY ---")
print(category_performance)

# Step 4: Sort categories by total revenue in descending order
sorted_categories = category_performance.sort_values(by="revenue",
ascending=False)
print("\n--- CATEGORIES RANKED BY REVENUE ---")
print(sorted_categories)

# Step 5: Customer Tier Frequency Distribution
tier_distribution = df["customer_tier"].value_counts()
print("\n--- TRANSACTION COUNT BY CUSTOMER TIER ---")
print(tier_distribution)
```

INTERPRETING NUMBERS & FORMULATING BUSINESS RECOMMENDATIONS

A. Observation (The Aggregated Output):

- Revenue Breakdown: Electronics generated $4,800.00 (68.4% of total revenue), followed by Apparel at $800.00, and Home at $720.00.
- Profit Margin Reality: While Electronics drives top-line revenue, Apparel maintains the highest average profit margin (43.3%), whereas Home has the lowest margin (12.3%).
- Order Distribution: Electronics accounts for 4 orders; Apparel and Home account for 3 orders each. Gold tier customers initiate the highest transaction frequency.

B. Interpretation & Business Meaning:

- Total stable revenue masks underlying structural divergence. Electronics is our volume and gross revenue engine, but Apparel is our high-margin efficiency driver.
- The Home category represents a risk: it generates low revenue ($720) and thin margins (12.3%), consuming operational overhead without proportional return.

C. Professional Recommendation to Marcus Thorne (Product Category Manager):

"An aggregated audit of our product catalog reveals that Electronics drives 68.4% of total revenue ($4,800), but Apparel delivers superior profitability with a 43.3% average profit margin. Conversely, the Home category underperforms across both volume ($720 total revenue) and margin (12.3%).

Recommendation: Protect and expand inventory for high-margin Apparel lines while conducting a pricing and supplier cost review for the Home category to either improve margins or reallocate floor space."

CORE MINDSET HABITS REINFORCED TODAY

1. Never stop at a total sum: Total revenue looked stable, but grouping exposed that low-margin product lines were dragging down overall profitability.
2. Balance volume with margin: High revenue does not automatically equal high profit; always evaluate margins alongside top-line figures.